

JEE ADVANCED CUMULATIVE TEST SERIES

44
YEARS
OF EXCELLENCE



NARAYANA
IIT-JEE ACADEMY-INDIA

Sec: SR.IIT_*CO-SC(MODEL-A)
Time: 3 Hrs

Date: 30-06-24
Max. Marks: 180

30-06-24_SR.IIT_STAR CO-SC(MODEL-A)_JEE ADV_REVIEW TEST-1_SYLLABUS

MATHEMATICS: (100% CUMULATIVE)
SYLLABUS COVERED FROM 22-04-23 TO 29-06-24

PHYSICS: (100% CUMULATIVE)
SYLLABUS COVERED FROM 22-04-23 TO 29-06-24

CHEMISTRY: (100% CUMULATIVE)
SYLLABUS COVERED FROM 22-04-23 TO 29-06-24

THE NARAYANA GROUP

TIME:3hrs**IMPORTANT INSTRUCTIONS****Max Marks: 180****MATHEMATICS**

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 4)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 5 – 7)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 8 – 13)	Questions with Non-Negative Integer type	+4	0	6	24
Sec – IV (Q.N : 14 – 17)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
Total				17	60

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 18 – 21)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 22 – 24)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 25 – 30)	Questions with Non-Negative Integer type	+4	0	6	24
Sec – IV (Q.N : 31 – 34)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
Total				17	60

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 35 – 38)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 39 – 41)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 42 – 47)	Questions with Non-Negative Integer type	+4	0	6	24
Sec – IV (Q.N : 48 – 51)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
Total				17	60

MATHEMATICS

MAX.MARKS: 60

SECTION-I (Maximum Marks: 12)

This section contains **FOUR (04)** questions.

Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer. For each question, choose the correct option corresponding to the correct answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If only the correct option is chosen.

Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)

Negative Marks: -1 In all other cases.

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined as $f(x) = [\sin x + x + 1]$,

Let $x_1, x_2, x_3, \dots, x_n$ are all the points where $f(x)$ is non-differentiable in

$$\left(-\frac{22}{7}, \frac{22}{7}\right),$$

$S = \sum_{i=1}^n x_i$, then which of the following is **CORRECT**?

(where $[x]$ represents greatest integer less than or equal to 'x')

- A) $S = 0$ B) $n = 6$ C) $S = 7$ D) $S + n = 13$

2. If the function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = x^3 - 3(a-2)x^2 + 3ax + 7$, for some $a \in \mathbb{R}$ is increasing in the interval $(0, 1]$ and decreasing in the interval $[1, 5)$.

Let (c, d) be the point on the curve $y = f(x)$ where tangent drawn to it does not meet the curve again at any other point, then value of $c + d$ is _____

- A) 6 B) 20 C) -20 D) 1

3. $f(a)$ be the number of distinct points in 2-D cartesian plane lying on both $(x-1)^2 + y^2 = 1$ and $|x-a| + |y| = |a|$ where $a \in \mathbb{R} - \{0\}$. Then which of the following options is/are **CORRECT** ?

- A) $f\left(\frac{5}{4}\right) = 6$ B) $f\left(\frac{5}{4}\right) = 2$ C) $f\left(\frac{5}{4}\right) = 5$ D) $f\left(\frac{11}{10}\right) = 7$

4. z_1 & z_2 are two non zero complex numbers in argand plane such that $|z_1 - z_2| = 4$, Imaginary part of $\frac{1}{z_1 \bar{z}_2}$ is zero and $\frac{1}{(z_1 - 1)(\bar{z}_2 - 1)}$ is purely imaginary, then maximum area of the triangle formed with vertices 1, z_1 and z_2 in argand plane is _____

- A) 1 B) 2 C) 4 D) 8

SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.

Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

5. If $A = \{(x, y, z) : x + 2y + 3z = 1, (x, y, z) \in \mathbb{R}^3\}$,

$$B = \{(x, y, z) : 2x + py + 6z = 2, (x, y, z) \in \mathbb{R}^3\},$$

$$C = \left\{ (x, y, z) : \frac{x}{3} + \frac{2y}{3} + qz = r, (x, y, z) \in \mathbb{R}^3 \right\}$$

Then which of the following is/are **CORRECT**?

A) If $p = 4, q = 1, r = \frac{1}{3}$, then $A \cap B \cap C$ is an Infinite set

B) If $p = 4, q = 1, r = 1$, then $A \cap B \cap C$ is a null set

C) If $p \neq 4, q = 1, r = 1/3$, then $A \cap B \cap C$ is a singleton set

D) If $p \neq 4, q \neq 1, r = 1/3$, then $A \cap B \cap C$ is a singleton set

6. Let A and B be two 3×3 matrices with real entries. X is any column matrix

Define $P = \{X : AX = O\}$ and $n(P) > 1$. Given $2A - 3I = (I - A)B$. $n(A_0)$ represents number of elements in the set ' A_0 '. Then which of the following is/are **CORRECT**?

A) If $R = \{X : AX = X\}$, then $n(R) = 1$

B) Let $Q = \left\{ X : AX = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$. If $\ell \in Q$, then for all $m \in P$, $\ell + m \in Q$

C) There exists a 3×3 non-zero matrix R such that $(B^2 + 2B)R$ is the zero matrix

D) $AB^2 - BA^2 \neq I$

7. Let $\{L_1, L_2, \dots, L_n\}$ be the set of all lines in 2-D Cartesian system such that L_i has only one common point with each of the curves $\frac{x^2}{4} + \frac{y^2}{1} = 1$ and $y^2 = 4(x - 2), \forall i = 1, 2, \dots, n$. Then which of the following is/are **CORRECT** ?
- A) $n = 5$
- B) $n = 3$
- C) Area of the figure bounded by all the lines $L_1, L_2, \dots, L_n$ is $5\sqrt{5}$ sq. Units
- D) Area of the figure bounded by all the lines $L_1, L_2, \dots, L_n$ is $10 - \sqrt{5}$ sq. Units

SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**

For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

8. Let $M = [a_{ij}]_{2 \times 2}$ is a matrix. Where

$$a_{ij} \in \{-2, -1, 0, 1, 2\}$$

$$A = \{M : M^2 = I\}$$

$$B = \{M : M^T = M\}$$

$$C = \{M : \text{Tr}(M) = 0\}$$

'I' represent identity matrix, X^T represents transpose of the matrix X, $\text{Tr}(X)$ represents sum of principal diagonal elements of the matrix 'X'. $n(X)$ represents number of elements in the set 'X'.

Then $n((A \cap B) \cup C)$ is equal to _____

9. If $\bar{a} = x_1 \bar{i} + y_1 \bar{j} + \bar{k}$, $\bar{b} = -\bar{i} + y_2 \bar{j}$, $\bar{c} = 2\bar{i} + x_3 \bar{j} + y_3 \bar{k}$

Such that $|\bar{a} - \bar{b}| = \sqrt{14}$, $|\bar{b} - \bar{c}| = \sqrt{56}$, $|\bar{c} - \bar{a}| = \sqrt{126}$. If $\bar{p} = \frac{\bar{a} + \bar{b}}{2}$, $\bar{q} = \frac{\bar{b} + \bar{c}}{2}$, $\bar{r} = \frac{\bar{c} + \bar{a}}{2}$

then the value of $|\bar{p} \times \bar{q} + \bar{q} \times \bar{r} + \bar{r} \times \bar{p}|$ is equal to _____

10. Let $a < b < c$ be positive integers which are in G.P. If $a + c = 2(b + 3)$ and there exists 'K' $\in \mathbb{Z}$ such that $b = Ka$, then value of $\left(\cot^{-1}(\cot b) - \frac{3}{2} \sin^{-1}(\sin a) \right)$ is

11. $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Let R_1 & R_2 be two relations defined on X such that

$$R_1 = \{(x, y) : x - y \text{ is divisible by } 3\}$$

$$R_2 = \{(x, y) : x - 2y \text{ is divisible by } 3\}$$

If $A \setminus B$ represents the set of all elements in A which are not in B and if $n(A)$ denotes the number of elements in the set A, then $n(R_1 \cup R_2) + n(R_1 \setminus R_2) =$ _____

12. $(x, y) \xrightarrow{R} (x_1, y_1)$ symbolically represents that the point (x, y) in 2-D cartesian plane gets transformed to (x_1, y_1) using operator R and if it is given that

$$(x, y) \xrightarrow{f_1} (y, x), (x, y) \xrightarrow{f_2} \left(\frac{x+y}{2}, \frac{x-y}{2} \right), (x, y) \xrightarrow{f_3} (-x, -y)$$

$\forall x, y \in \mathbb{R}$. If $(\sin \theta, \cos \theta)$ where $\theta \in \mathbb{R}$ is transformed as shown below

$$(\sin \theta, \cos \theta) \xrightarrow{f_3} (A, B) \xrightarrow{f_2} (C, D) \xrightarrow{f_1} (E, F)$$

$$(\sin \theta, \cos \theta) \xrightarrow{f_1} (G, H) \xrightarrow{f_2} (I, J) \xrightarrow{f_3} (K, L) \xrightarrow{f_1} (M, N)$$

Then the square of maximum possible distance between (E, F) & (M, N) is

13. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable in $\mathbb{R}$ and it satisfies

$$xf(x) + \int_0^x f(t) dt = e^x (x \cos x + x \sin x + \sin x) \quad \forall x \in \mathbb{R}. \text{ The number of times } f(x) \text{ vanishes in the interval } [-2024\pi, 2024\pi] \text{ is } \underline{\hspace{2cm}}$$

SECTION - IV (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.

The answer to each question is a **NUMERICAL VALUE**.

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal places.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

Paragraph-I for Question Nos. 14 and 15:

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as $f(x) = \int_{-1}^x \left(\sin(t-3) + t^2 - 6t + \frac{11}{3} \right) dt$

14. The maximum value of x for which $f(x)$ becomes zero is

15. The number of integral values of x in $[8, 16]$ for which $f(x)$ is positive is

Paragraph-II for Question Nos. 16 and 17:

Consider the functions $f, h : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \sin^{-1} \left(\frac{4x}{4+x^2} \right)$

and $h(x) = \frac{x^2 + x + 1}{2}$. Let $\lambda = f(\sqrt{7} - 2) - f\left(\frac{4}{\sqrt{7} - 2}\right)$ and 'm' represents the slope of

tangent to $y = h(x)$ at $x = \lambda$. Let $g : [0, 1] \rightarrow \mathbb{R}$ be defined by $g(x) = 2^{\alpha x} + 2^{\alpha(1-x)}$

where $\alpha = \sum_{k=1}^{\infty} m^{2k}$,

16. Minimum value of $g(x)$ is m , value of $\log_2 m$ is _____

17. Maximum value of $g(x)$ is M , value of $\log_2 (M-1)$ is _____

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PHYSICS

MAX.MARKS: 60

SECTION-I (Maximum Marks: 12)

This section contains **FOUR (04)** questions.

Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer. For each question, choose the correct option corresponding to the correct answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If only the correct option is chosen.

Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)

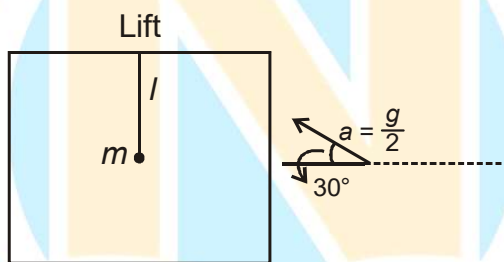
Negative Marks: -1 In all other cases.

18. The pressure caused due to gravitational attraction, inside a solid sphere of density ρ ,

radius R , at a distance $r = \frac{R}{2}$ from the centre of sphere, is

- A) $\frac{\pi G \rho^2 R^2}{4}$ B) $\frac{\pi G \rho^2 R^2}{8}$ C) $\frac{\pi G \rho^2 R^2}{2}$ D) $\frac{3\pi G \rho^2 R^2}{8}$

19. A simple pendulum is suspended from the ceiling of an elevator which is being accelerated with $a = \frac{g}{2}$ as shown in figure. The time period of small oscillations of the simple pendulum around its mean position w.r.t the elevator is



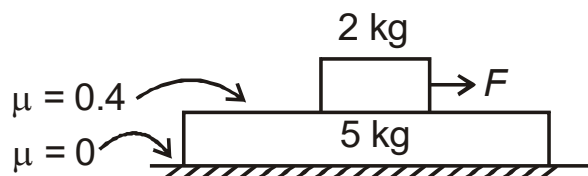
- A) $2\pi \sqrt{\frac{l}{g} \sqrt{\frac{2}{5}}}$ B) $2\pi \sqrt{\frac{l}{g} \sqrt{\frac{3}{2}}}$ C) $2\pi \sqrt{\frac{l}{g\sqrt{3}}}$ D) $2\pi \sqrt{\frac{2l}{g\sqrt{7}}}$

20. Consider a tube of length l , radius R in which a liquid of density ρ is flowing steadily. The velocity of liquid in the tube is given by $v = v_0 \left(1 - \frac{r^2}{R^2}\right)$, where r is the radial distance from the axis and v_0 is constant. The kinetic energy of liquid in tube's volume is given by (Take, $l = 6R$)

- A) $\frac{2}{3} \pi \rho v_0^2 R^3$ B) $\pi \rho v_0^2 R^3$ C) $\frac{\pi \rho v_0^2 R^3}{6}$ D) $\frac{\pi \rho v_0^2 R^3}{3}$

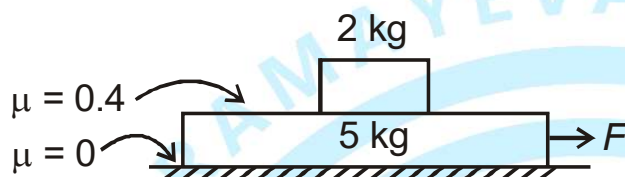
21. Select the value of $F_C + F_A + F_B + F_D$. ($g = 10 \text{ m/s}^2$, $\sqrt{3} = 1.73$)

(A)



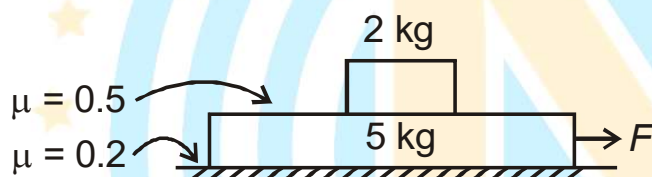
The applied force F is just sufficient to cause slipping between the 2 kg and 5 kg block. Call it F_A

(B)



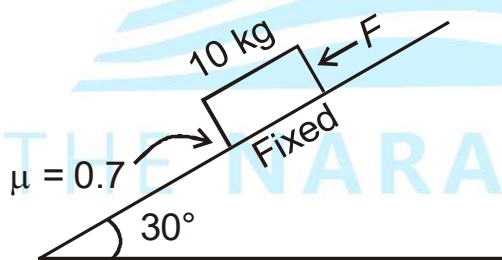
The applied force F is just sufficient to cause slipping between the 2 kg and 5 kg block. Call it F_B

(C)



The applied force F is just sufficient to cause slipping between the 2 kg and 5 kg block. Call it F_C

(D)



The applied force F is just sufficient to cause slipping between the block and inclined surface. Call it F_D

A) 98.75

B) 96.75

C) 100.37

D) 95.25

SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.

Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

22. Select the correct statement(S) related to specified thermodynamic processes on a gas.

A) Mono-atomic gas obeys $TV = \text{constant}$. The molar heat capacity of the gas in the process is $1.5R$

B) One mole of di-atomic gas undergoes a process that obeys $PV^{\frac{3}{2}} = \text{constant}$. The work done on the gas while temperature changes from T to $2T$ is $2RT$

C) 4 moles of diatomic gas obeys $PT = \text{constant}$. The heat absorbed by gas while its temperature increases from T to $2T$ is $18RT$

D) Monoatomic gas is heated such that amount of heat supplied to it is equal to decrease in internal energy of the gas. The work done on gas when 1 mole gas temperature is increased from T to $2T$ is $-3RT$

23. A charged particle of charge q , mass m is projected with a velocity v in a region of uniform electric and magnetic fields E and B respectively, such that v , E and B are mutually perpendicular to each other and $v = \frac{E}{B}$. Neglect gravity. Then

A) If E is switched off, then particle will follow a circular path

B) If B is switched off, then particle will follow a parabolic path

C) If E is switched off, for a time of $\frac{\pi m}{4qB}$ and then switched on, the path will not be straight line

D) If E is switched off, for a time of $\frac{2\pi m}{qB}$ and then switched on, the path of particle will be a straight line

24. Given an LC circuit with the initial charge on the capacitor as $Q_0 = 5 \mu C$ and initial current in the inductor as zero. The circuit parameters are $C = 400 \mu F$ and $L = 90 \text{ mH}$. Determine the following parameters:

- 1) Maximum voltage across the inductor
- 2) Maximum current in the inductor
- 3) Maximum energy stored in the inductor
- 4) Charge on the capacitor when the current in the inductor is half of its maximum value

Choose the correct answers (one or more) from the options below:

Options:

- A) $1.25 \times 10^{-2} \text{ V}$ B) $8.33 \times 10^{-4} \text{ A}$ C) $3.13 \times 10^{-8} \text{ J}$ D) $4.33 \times 10^{-6} \text{ C}$

SECTION-III (Maximum Marks : 24)

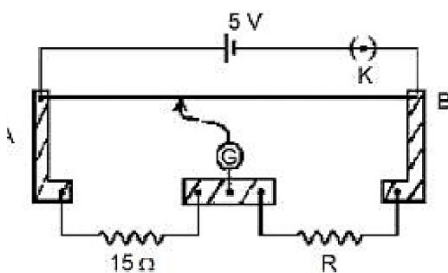
This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**. For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.

Answer to each question will be evaluated according to the following marking scheme:

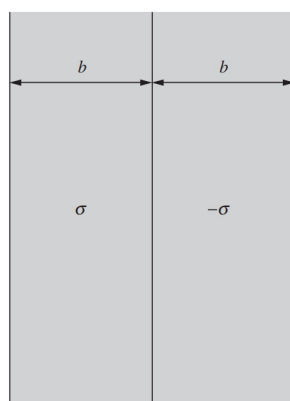
Full Marks: +4 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

25. A meter bridge setup is shown in the figure. It is used to determine an unknown resistance R using a given resistor of 15Ω . The galvanometer (G) shows null deflection when the tapping key is at the 43 cm mark from end A. End corrections account for the additional resistance introduced due to the connecting wires and terminals, which are not part of the meter bridge wire itself. This correction ensures accurate measurements of unknown resistances. In this experiment, the end correction for end A is 2 cm and none at B. Given this information, calculate the determined value of R (in ohm).



26. A thin prism of angle 6° and refractive index for yellow light $n_Y = 1.5$ is combined with another prism of angle 5° and $n_Y = 1.55$. The combination produces no dispersion. The net average deviation (δ) produced by the combination is $\left(\frac{1}{x}\right)^\circ$. The value of x is _____.
27. A travelling microscope is used to determine the refractive index of a glass slab. If 40 divisions are there in 1 cm on the main scale and 50 Vernier scale divisions are equal to 49 main scale divisions, then the least count of the travelling microscope is _____ $\times 10^{-6}$ m.
28. A uniform solid disk with mass 4 kg and radius 1 meter rotates about an axis perpendicular to its face and passing through its center. A thin rod of negligible mass is attached to the center of the disk and extends outwards. A small sphere with mass 0.2 kg is fixed to the free end of the rod at a distance 0.5 meters from the center of the disk. An external torque of 8.2 Nm is applied about the rotation axis for a duration of 2 seconds. If we define the final angular velocity ω of the system as Ω , what is 50 times the value of Ω ?
29. Two thin strips of charge are placed parallel to each other with their centers separated by a distance b . The left strip has a positive surface charge density $\sigma = 500 \mu\text{C} / \text{m}^2$ and the right strip has a negative surface charge density $-\sigma = -500 \mu\text{C} / \text{m}^2$. The width of each strip is also $b = 0.2\text{m}$, and the strips are infinitely tall.



Given:

Surface charge density, $\sigma = 500 \mu\text{C} / \text{m}^2 = 500 \times 10^{-6} \text{C} / \text{m}^2$

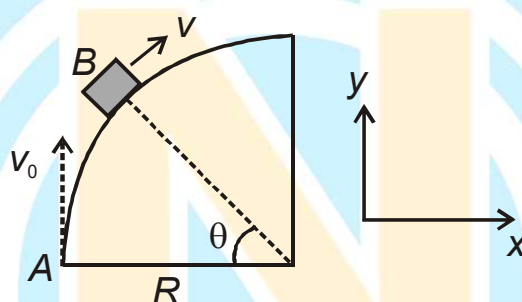
Width of each strip, $b = 0.2\text{m}$

Coulomb constant, $k = 9 \times 10^9 \text{Nm}^2 / \text{C}^2$

The natural logarithm value, $\ln(2) = 0.7$

Determine the force per unit height (in N/m) between the two strips.

30. A particle starts from point A and moves upon circular path such that the x – component of the acceleration is always constant in magnitude. If at point B , the speed is $v_0 \frac{2}{\sqrt{3}}$. Find angle θ in degrees.



SECTION - IV (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.

The answer to each question is a **NUMERICAL VALUE**.

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal places.

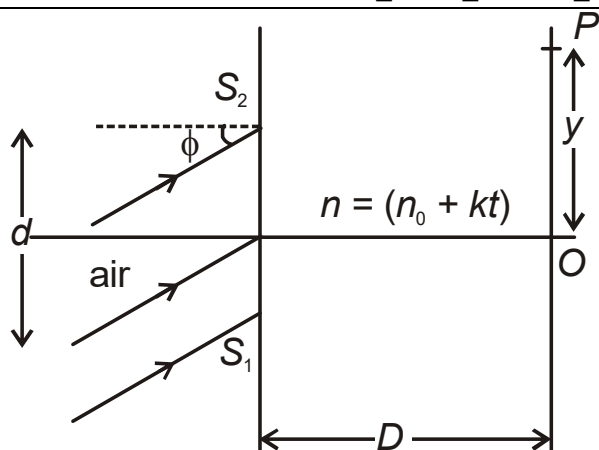
Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

Paragraph-I for Question Nos. 31 and 32:

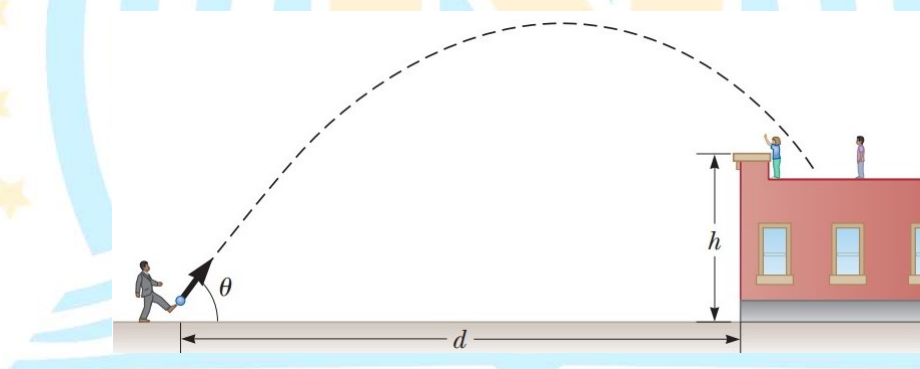
A monochromatic source of light with wavelength $\lambda = 600 \times 10^{-9} \text{m}$ is used in a Young's double slit experiment. The light strikes the slits S_1 and S_2 in the air medium, which are separated by a distance $d = 0.5 \text{mm}$. The medium between the plane of the slits and the screen has a refractive index that varies with time as $n = n_0 + kt$ with $n_0 = 1.5$ and $k = 0.02 \text{s}^{-1}$. The distance between the slits and the screen is $D = 2 \text{m}$, and the angle $\phi = 0.01$ radians.



31. Determine the y-coordinate (in centimeters) of the central maxima at a time $t = 5$ s.
32. Find the velocity of the central maxima (in mm / s) as a function of time at $t = 5$ s.

Paragraph-II for Question Nos. 33 and 34:

A playground is on the flat roof of a city school, 6.00 m above the street below. The vertical wall of the building is $h=7.00$ m high, forming a 1-m-high railing around the playground. A ball has fallen to the street below, and a passerby returns it by launching it at an angle of $\theta = 53.0^\circ$ above the horizontal at a point $d = 24.0$ m from the base of the building wall. The ball takes 2.20 s to reach a point vertically above the wall. Take $g = 9.8 \text{ m/s}^2$. $\cos 53^\circ = 0.6$.



33. Determine the horizontal distance (in meters) from the wall to the point on the roof where the ball lands.
34. Find the speed of the ball (in meters per second) when it reaches the point vertically above the wall.

CHEMISTRY**MAX.MARKS: 60****SECTION-I (Maximum Marks: 12)**

This section contains **FOUR (04)** questions.

Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer. For each question, choose the correct option corresponding to the correct answer.

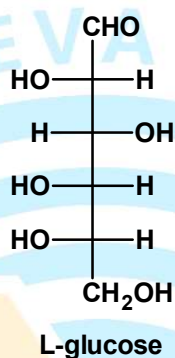
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If only the correct option is chosen.

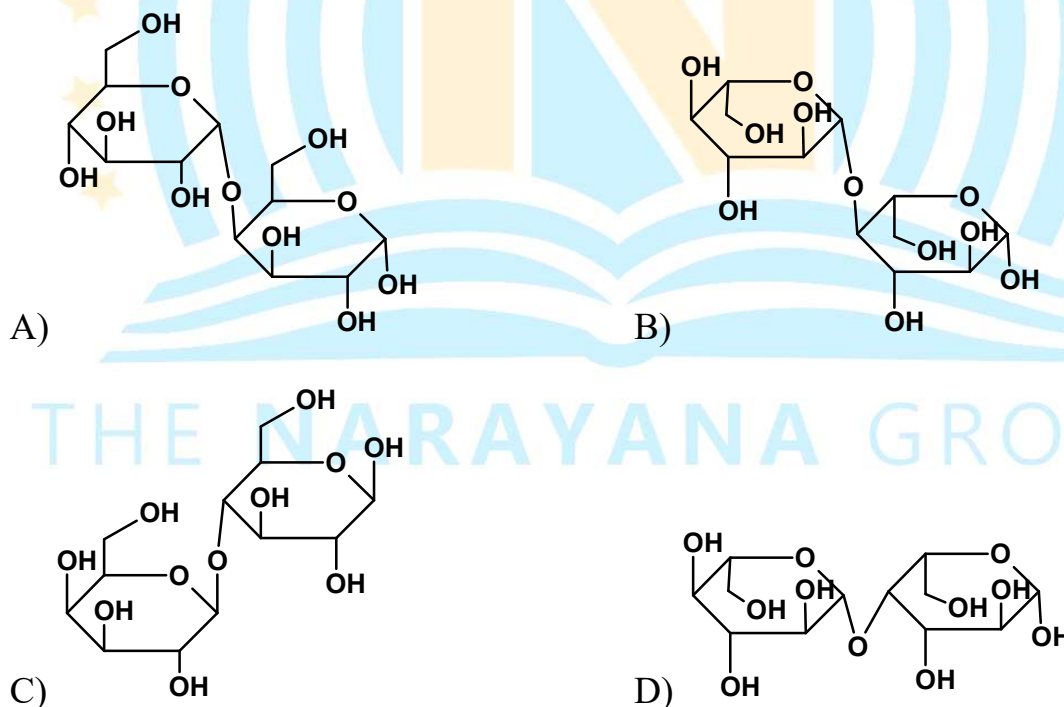
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)

Negative Marks: -1 In all other cases.

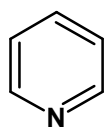
35. Lactose is a disaccharide of β -D-galactose and β -D-glucose linked through C₁-C₄ glycosidic linkage. Galactose is C₄-epimer of glucose.



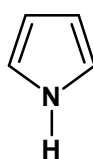
Based on the information given, which of the following correctly represents D-Lactose?



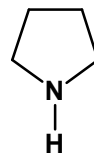
36. The correct order of basic strength of following compounds (I – IV) is:



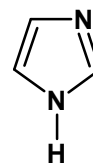
[I]



[II]



[III]



[IV]

A) [II] < [I] < [IV] < [III]

B) [IV] < [I] < [II] < [III]

C) [II] < [I] < [III] < [IV]

D) [I] < [II] < [III] < [IV]

37. The total number of lone pairs of electrons in hypophosphite and chlorite ions, respectively, is:

A) 7, 7

B) 5, 7

C) 5, 5

D) 7, 5

38. Atom X occupies fcc lattice sites as well alternate tetrahedral voids of the same lattice. The packing efficiency (in %) of the resultant solid is close to:

A) 25

B) 35

C) 55

D) 75

SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.

Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

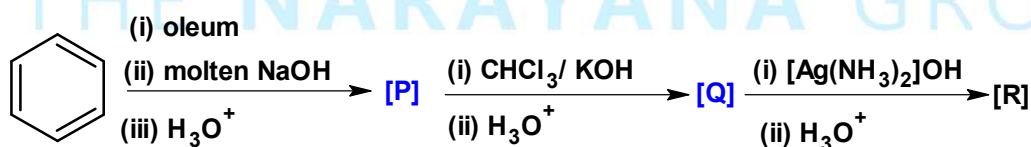
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

39. In the following reaction sequence, [P], [Q] and [R] are the major products.



Choose the correct option(s)

A) [P] and [Q] gives violet coloration with neutral FeCl_3 B) [P] and [R] gives white precipitate with $\text{Br}_2 / \text{H}_2\text{O}$ C) [Q] on reaction with $\text{Zn-Hg} / \text{HCl}$ gives catecholD) [R] on reaction with $(\text{CH}_3\text{CO})_2\text{O} / \text{H}^+$ gives 2-acetylbenzoic acid.

40. The correct statement(s) regarding colloids is(are):
- A) Brownian motion of colloidal particles does not depend on the size of the particles but depends on viscosity of solution
 - B) Tyndall effect is observed when the size of sol particle is much smaller than wavelength of light used
 - C) The process of precipitating colloidal sol by an electrolyte is called peptization
 - D) A negatively charged colloid is obtained upon addition of limited AgNO_3 to aqueous KI.
41. Among the following, the option(s) which involve a redox change is(are):
- A) Addition of copper powder to aqueous CuCl_2
 - B) Heating Pb_3O_4 with dilute HNO_3 to obtain PbO_2
 - C) Passing SO_2 gas into aqueous FeCl_3 solution
 - D) Passing Cl_2O gas into hot NaOH solution

SECTION-III (Maximum Marks : 24)

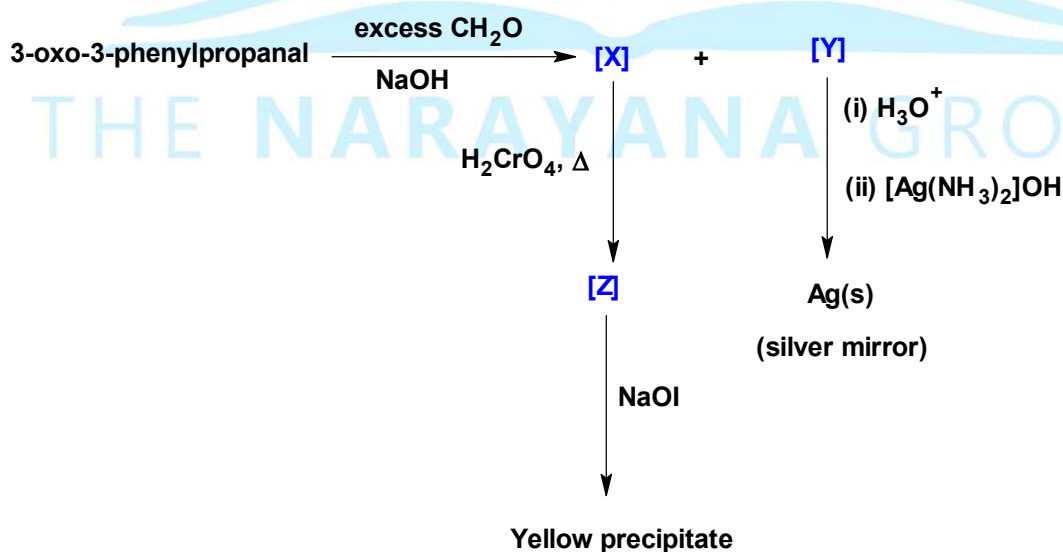
This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**. For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: ★ +4 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

42. Consider the following reaction sequence



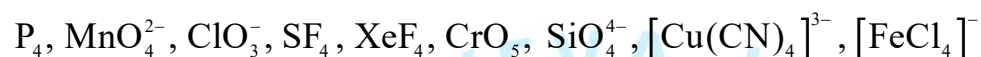
The number of moles of CH_2O consumed in the reaction is 'x' and

The molar mass of [Z] is 'y' g/mol

The value of x.y (in g) is _____.

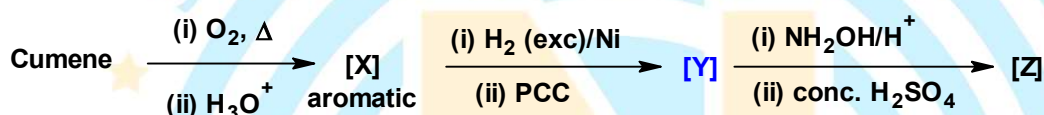
[Given, atomic weight: H = 1, C = 12, N = 14, O = 16, Na = 23, Ag = 108, I = 127]

43. Among the following, the number of diamagnetic species having tetrahedral shape is ____.



[Given, atomic number: O=8, F=9, Si=14, P=15, S=16, Cl=17, Cr=24, Mn=25, Fe=26, Cu=29, Xe=54]

44. Consider the following reaction sequence:

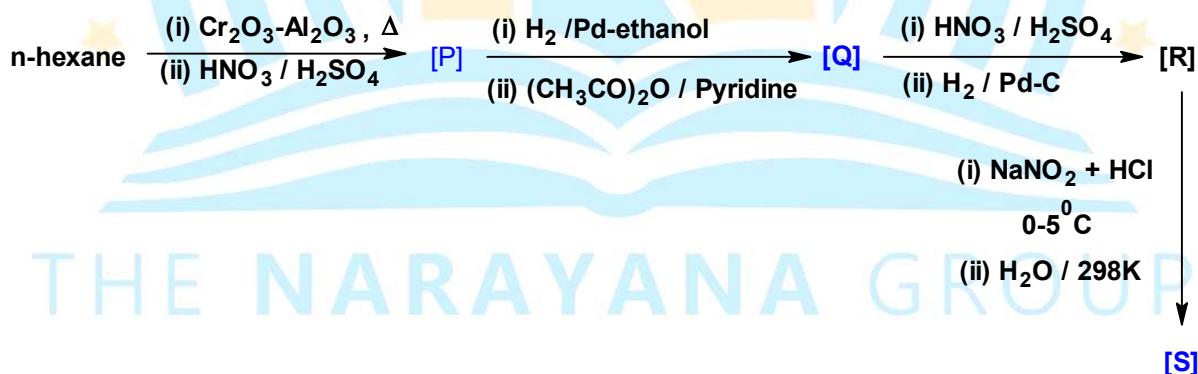


The percentage of carbon in [Z] is _____.

[Given, atomic weight: H = 1, C = 12, N = 14, O = 16, S = 32]

(Round off to nearest integer)

45. Consider the following reaction sequence:



The total number of lone pairs of electrons involved in delocalization in [S] is _____.

46. One mole of a monoatomic gas confined in 25L flask at 300K exert a pressure of 0.95 atm, whereas expected pressure was 1.0 atm has the gas behaved ideally. The Boyle's temperature (K) of the gas is _____.

[Given, R = 0.08 Latm /mol K]

(Round off to nearest integer)

47. Phenol undergoes partial dimerisation in CHBr_3 solvent. When 7.52g of phenol is added to 100g of CHBr_3 , the depression of freezing point observed to be 7K. If K_f of CHBr_3 is 14K kg mol^{-1} , the percentage dimerisation of phenol in bromoform is _____.

[Given, atomic weight: $\text{H} = 1$, $\text{C} = 12$, $\text{O} = 16$, $\text{Br} = 80$]

SECTION - IV (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.

The answer to each question is a **NUMERICAL VALUE**.

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal places.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

Paragraph-I for Question Nos. 48 and 49:

0.5g of a weak base (BOH) present in 50mL solution is titrated with 0.1M HCl. The pH at half equivalence and equivalence points are 9 and 5.15 respectively. The molar conductivity of BOH at infinite dilution is $400\text{ Scm}^2\text{ mol}^{-1}$.

48. The molar mass (g/mol) of BOH is _____.
49. The conductivity (Sm^{-1}) of initial BOH solution is _____.

Paragraph-II for Question Nos. 50 and 51:

4.96g white phosphorus on reaction with SO_2Cl_2 forms [Q] and [R]. [Q] is non polar whereas [R] is polar. Half of the amount of [Q] is reacted with silver powder to form [T] and [U] and remaining half are reacted with tin powder to form [V] and [U].

[Atomic weight: $\text{O} = 16$, $\text{P} = 31$, $\text{S} = 32$, $\text{Cl} = 35.5$, $\text{Ag} = 108$, $\text{Sn} = 119$]

50. The weight (g) of [T] formed is _____.
51. The weight (g) of [V] formed is _____.